

Stock-AI: AI Integrated Stock Trading Platform Startup Instructions

Spring 2026

COM S 4020 - UG_HF_03

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1. Database Deployment

- a. Establish relational database service
 - i. Must be MySQL or MySQL compatible (e.g., MariaDB).
 - ii. Make note of the hostname of the running service's
 - iii. Make note of the port number it's running on; typically 3306.
- b. Create a database instance
 - i. Make note of the name, as this will house all the tables required for Stock-AI.
- c. Create database user
 - i. Make sure it has permissions over the newly created database instance.
 - ii. Make note of the username and password; these will need to be populated into the backend code.
- d. Ensure that the relational database service is accessible to deployment location of the backend; make note of hostname and port number of the database service.
- e. Do not set up any database schema, as this is automatically handled by the backend code.

2. FastAPI Backend Deployment

- a. Place files in desired location
 - i. Place the Stock-AI codebase in whatever deployment location on your deployment.
 - ii. Navigate to /Backend/FastAPI/
 - iii. Update credentials in .env file
 1. Open /Backend/FastAPI/.env
 2. Replace <YOUR-DB-USER> with the user created above
 3. Replace <YOUR-DB-USER-PASSWORD> with the password you set
 4. Replace <YOUR-DB-HOSTNAME> with your host
 5. Replace <YOUR-DB-PORT> with the port the database service is running on; this is usually 3306 by default.
 6. Replace <ANY-SECRET-KEY> with any string.
 - a. Note: this is used for security purposes. Changing this later will break users' ability to log in, as this key is needed to match passwords.
 7. Replace FINNHUB-KEY with your Finnhub API key.
 - a. Obtain for free (as of Spring 2026) from here: <https://finnhub.io/>
 8. Replace CRON_SECRET with any string.

- a. This is used as an authorization measure for a separate automated system to call an internal route for daily tasks (value tracking). Our team had this set up on GitLab, but it is not necessary for first time setup.
 - 9. Replace OPENAI_API_KEY with your OpenAI key.
 - a. This will require an account with OpenAI.
- iv. Build & Run: 2 options
 - 1. **Recommended:** build & run inside container
 - a. Docker
 - i. Run "Docker compose up --build"
 - ii. Application should automatically build and run on port 8080 of host machine
 - b. Podman
 - i. Build
 - 1. Run "sudo podman build -t myfastapi ."
 - ii. Run Container
 - 1. Run "sudo podman run -d --name stock-ai-backend --env-file .env -p 8080:8080 myfastapi"
 - 2. Application should now be running on port 8080 of host machine
 - 2. Python virtual environment
 - a. This is less recommended because there may be some dependency issues to resolve; for reference, our backend application was built using Python 3.12.
 - b. **Optional:** Clear old environment
 - i. Windows: run "rmdir /s /q venv"
 - ii. Linux: run "rm -rf venv"
 - c. Create virtual environment
 - i. Windows: run "python -m venv venv"
 - ii. Linux: run "Python3 -m venv venv"
 - d. Start Python virtual environment
 - i. Windows: run "venv\Scripts\activate"
 - ii. Linux: run "source venv/bin/activate"
 - e. Install dependent packages
 - i. Run "pip install --no-cache-dir -r requirements.txt"

- ii. Note: If adding new packages in the future, use “pip freeze > requirements.txt” to save new dependencies
 - f. Run application
 - i. Run “uvicorn app.main:app --host 0.0.0.0 --port 8080”
 - ii. Application should now be running on port 8080 of host machine
 - v. Verify running status
 - 1. In a browser, head to `http://<YOUR-HOST>:8080/docs`, and the automatically generated Swagger UI docs should be available.

3. React Frontend Deployment

- a. Update Backend URL
 - i. Navigate Frontend/src/types/[api.ts](#)
 - ii. Replace <YOUR-HOSTNAME> and <YOUR-PORTNUMBER> with your backend hostname and portnumber in const api.
- b. To deploy frontend:
 - i. In the frontend root directory:
 - 1. Run “npm run build”
 - 2. A folder named “build” or “dist” will be created
 - 3. Copy contents of folder into desired location (for example, /var/www/html)
- c. To run frontend locally:
 - i. In the frontend root directory:
 - 1. Run “npm run dev”
 - 2. The website will now be accessible via localhost on port 5173 by default.